Synthesis of Ethyl Caprylate via Fischer Esterification

Aidan Leonard, Dr. Kawagoe, Dr. MacPherson

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1 Introduction

The objective of this experiment is to synthesize ethyl caprylate via Fischer Esterification by reacting absolute ethanol with caprylic acid in the presence of catalytic concentrated sulfuric acid. The product will be characterized and analyzed by odor analysis, ATR-IR, 100 MHz ^{1}H NMR, and GC-MS.

2 Reaction Scheme

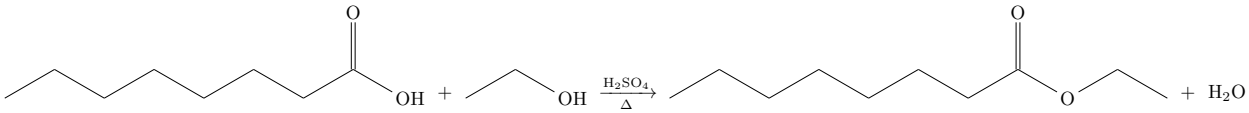


Figure 1: Reaction of Caprylic Acid and Ethanol

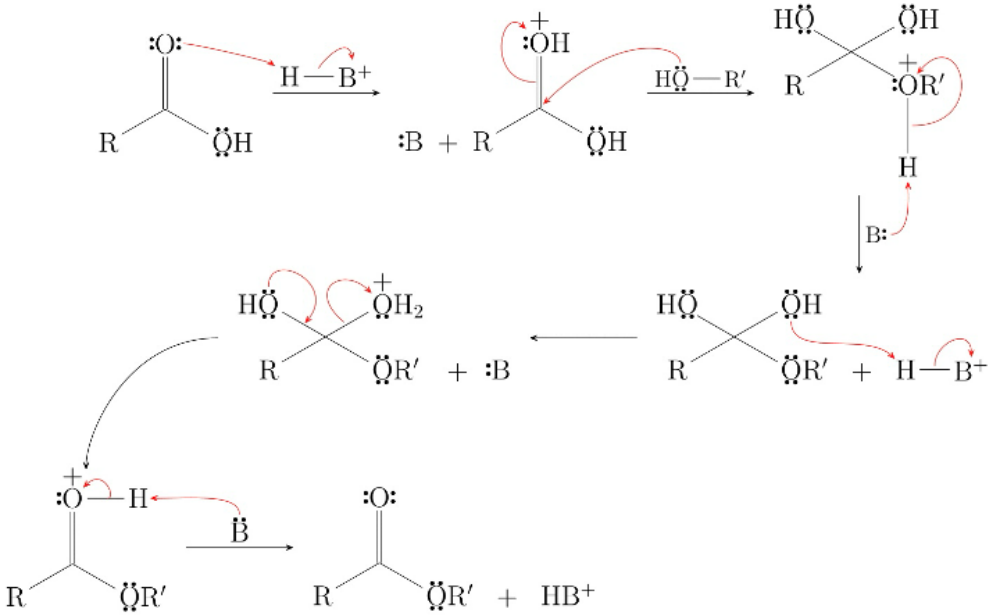


Figure 2: Fischer Esterification Mechanism

3 Table of Reagents and Products

Compound	MW (g/mol)	Density (g/mL)	B.P (°C)	Equiv	Amount	mmol
Caprylic Acid	144.21	0.91	237 - 239.7	1.0	7.20 g / 7.91 mL	49.9
Ethanol	46.07	0.789	78.2 - 78.4	3.0	6.90 g / 8.74 mL	149.8
Sulfuric Acid	98.08	1.84	–	cat.	5 drops	–
Diethyl Ether	74.12	0.710	34.6	solv.	–	–
Ethyl Caprylate	172.27	0.86 - 0.87	206 - 208	–	–	–

4 Experimental Procedure

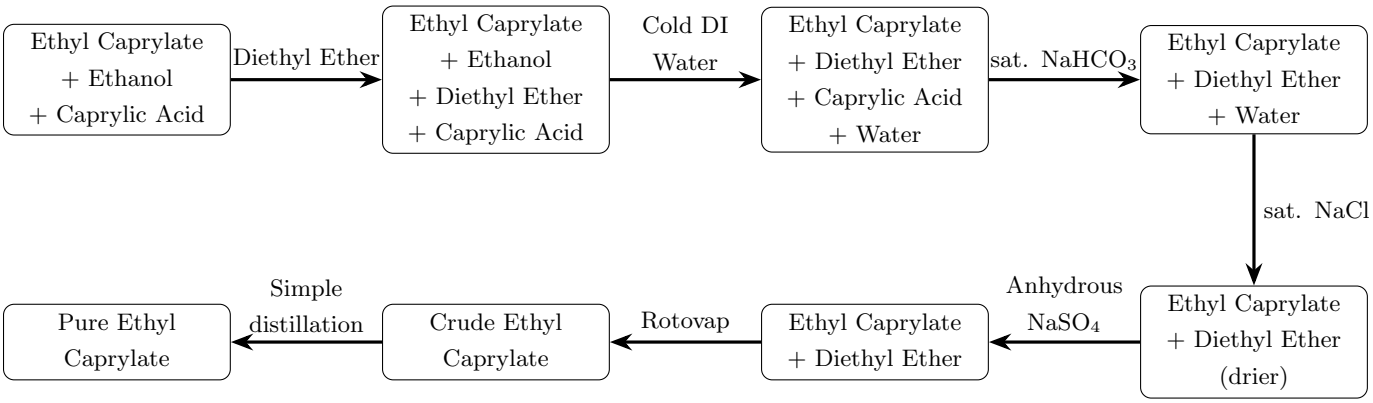
4.1 Apparatus

- Reflux
- Simple Distillation
- Rotovap

4.2 Procedure

- In a 25 mL round-bottom flask, add 7.20 g (7.91 mL) of caprylic acid followed by 6.90 g (8.74 mL) of absolute ethanol and a stir bar.
- While stirring, slowly add 5 drops of concentrated sulfuric acid.
- Gently reflux for 60 minutes, then allow the mixture to cool to room temperature.
- Chill the product in an ice bath then transfer the mixture to a separatory funnel using 20 mL total of diethyl ether to aid in the process.
- Quench the organic layer with 15 mL of iced cold DI water and collect the organic layer. Do this twice.
- Neutralize and extract the organic layer with small portions of a saturated sodium bicarbonate solution until the production of CO_2 gas ceases.
- Wash the remaining organic layer with a brine solution then dry the organic over anhydrous sodium sulfate followed by gravity filtration.
- Gently boil off the ether solution in a 50 mL beaker blowing a light stream of air over the top to obtain the crude ethyl caprylate.
- Purify the product with simple distillation.
- Analyze the purified product by odor assessment, ATR-IR, ^{1}H NMR (100 MHz), and GC-MS to confirm ester formation and purity.

5 Purification Scheme



6 Results and Data

g of Caprylic acid	g of Ethanol	Drops of H_2SO_4	Reflux Time
Mass of Crude Product		Mass of Purified Product	
Theoretical Yield			

Crude Percent Yield	Pure Percent Yield
Crude Product Observations	Purified Product Observations